



Week 4 Solution

Question 1

Question

A context switch occurs when a state of process changes from **New to Ready**.

State True or False.

Answer

False

Solution

A **context switch** occurs when the CPU switches execution from one process or thread to another.

When a process moves from **New** → **Ready**, it has just been created and placed in the ready queue. This transition does not necessarily mean that the CPU switches from one executing process to this new process.

A context switch generally involves saving the context of the currently running process and loading the context of another process.

Therefore:

New → Ready

does not by itself cause a context switch.

Correct Answer: False

Question 2

Question

A thermal sensor in the CPU checks the temperature of the machine. If it is more than 40°C, an interrupt is generated to shut down the machine.

This interrupt is a hardware interrupt.

State True or False.

Answer

True

Solution

A **hardware interrupt** is generated by a hardware device or hardware event that requires immediate attention from the CPU.

In this case:

```
graph TD; A[Thermal Sensor] --> B[Temperature exceeds limit]; B --> C[Hardware interrupt generated]; C --> D[CPU handles interrupt]; D --> E[Machine shutdown / protection action];
```

Since the thermal sensor is a hardware component generating the interrupt, this is classified as a **hardware interrupt**.

Therefore:

Correct Answer: True

Question 3

Question

What is the correct order of the following events?

- I. Device raises interrupt
- II. CPU checks signal on INT pin and sends INT ACK signal
- III. Interrupt Handler Routine is executed
- IV. Interrupt Controller sends INT signal to CPU
- V. CPU identifies which device has raised the interrupt

Options:

- 1. I, II, III, IV, V
- 2. V, I, IV, II, III
- 3. I, IV, II, V, III
- 4. II, IV, I, V, III

Answer

I, IV, II, V, III

Solution

The interrupt-handling process occurs in the following order:

Step I: Device raises an interrupt

A hardware device requires attention.

Device

↓

Interrupt request

Step IV: Interrupt Controller sends the INT signal to the CPU

The interrupt controller receives/manages the interrupt request and signals the CPU.

```
Device
  ↓
Interrupt Controller
  ↓
INT signal
  ↓
CPU
```

Step II: CPU checks the INT pin and sends INT ACK

The CPU detects the interrupt signal and acknowledges it.

Step V: CPU identifies the interrupt source

The CPU determines which device or interrupt source generated the request.

Step III: Interrupt Handler Routine is executed

Finally, the corresponding **Interrupt Handler Routine (ISR)** is executed.

Therefore, the complete sequence is:

```
I → IV → II → V → III
```

Correct Answer: I, IV, II, V, III

Question 4

Question

I/O APICs are used to handle interrupts from thermal sensors and internal timers.

State True or False.

Answer

False

Solution

The assignment distinguishes interrupts from external I/O devices and interrupts generated by internal processor components.

An **I/O APIC** is associated with handling and routing interrupts from I/O devices.

Thermal sensors and internal timers are internal sources rather than ordinary external I/O devices.

Therefore, the statement is:

False.

Question 5

Question

Match the following x86 interrupt-vector mnemonics with the type of interrupt:

Mnemonic	Type
I. #0F	a. Trap

Mnemonic	Type
II. #DF	b. Fault
III. #SS	c. Abort

Options:

1. I-a, II-b, III-c
2. I-a, II-c, III-b
3. I-b, II-c, III-a
4. I-c, II-b, III-a

Answer

I-a, II-c, III-b

Solution

The correct classification is:

#0F — Trap

#0F represents the **Overflow** exception.

It is classified as a:

Trap

#DF — Abort

#DF represents a **Double Fault**.

It is classified as an:

Abort

#SS — Fault

#SS represents a **Stack-Segment Fault**.

It is classified as a:

Fault

Therefore:

```
#0F → Trap  
#DF → Abort  
#SS → Fault
```

Hence:

Correct Answer: I-a, II-c, III-b

Question 6

Question

A CPU has four processes executing and uses a non-preemptive scheduler.

What is the total number of possible schedules for the 4 processes?

Answer

24

Solution

With four processes:

```
P1, P2, P3, P4
```

A non-preemptive schedule means that once a process starts executing, it continues until it finishes or voluntarily gives up the CPU.

The number of possible orders is the number of permutations of 4 processes:

$$4!$$

Therefore:

$$\begin{aligned} 4! &= 4 \times 3 \times 2 \times 1 \\ &= 24 \end{aligned}$$

Hence, the total number of possible schedules is:

24

Question 7

Question

More than one system call in an operating system can have the same system call number.

State True or False.

Answer

False

Solution

A system call number is used by the operating system to identify which system service the program is requesting.

Conceptually:

System Call Number



System Call Table



Corresponding System Call Handler

Each system call number must uniquely identify the appropriate system call within a given system-call interface.

If two different system calls had the same number, the operating system would not be able to determine which one should be executed based only on that number.

Therefore:

Correct Answer: False

Question 8

Question

Parameters in system calls are passed using user-mode stacks in Linux and by using registers in xv6.

State True or False.

Answer

False

Solution

The statement claims that:

Linux → User-mode stacks
xv6 → Registers

The assignment marks this statement as **False**, so this exact comparison is not correct.

The mechanism used for passing system-call information depends on the operating system implementation and architecture. Therefore, the statement should not be accepted as a correct general description.

Hence:

Correct Answer: False

Question 9

Question

Read the following statements and identify which are True or False.

- I. A context switch can occur when a process moves from a **Blocked to Ready** state.
- II. A context switch can occur when a process moves from a **Running to Zombie** state.
- III. A context switch can occur when a process moves from a **Running to Ready** state.

Answer

I - False, II - True, III - True

Solution

Statement I: **Blocked → Ready**

False

When a blocked process becomes ready, it is added to the ready queue.

This transition alone does not necessarily mean that the CPU switches to that process.

Blocked → Ready

The currently running process may continue executing.

Therefore, this transition does not by itself imply a context switch.

Statement II: Running → Zombie

True

When a running process terminates, it enters the zombie state until its exit status is collected.

Since the process is no longer executing, the CPU must switch to another process.

Therefore, a context switch can occur.

```
Running
  ↓
Zombie
  ↓
Another process scheduled
```

Statement III: Running → Ready

True

When a running process is preempted, it moves back to the ready state.

The CPU can then be assigned to another ready process.

```
Running
  ↓
Preemption
  ↓
Ready
```

This involves switching execution to another process.

Therefore, a context switch can occur.

Hence:

```
I    → False
II   → True
III  → True
```

Correct Answer: I - False, II - True, III - True

Question 10

Question

Timer interrupt latency should be as high as possible for smooth context switching.

State True or False.

Answer

False

Solution

Interrupt latency is the time between an interrupt occurring and the CPU beginning to handle it.

For a timer interrupt:

Timer Interrupt Occurs

|

| Interrupt latency

↓

CPU starts handling interrupt

A high interrupt latency means the CPU takes longer to respond.

For efficient scheduling and smooth context switching, the operating system generally requires a quick response to timer interrupts.

Therefore, interrupt latency should be **low**, not as high as possible.

Hence:

Correct Answer: False